MAT 319 HW03

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Problem 10.1. *Which of the following sequences are increasing? decreasing? bounded?*

(a) $1/n$

(b) $(-1)^n/n$

(c) n^5

(d) $\sin(n\pi/7)$

(e) $(-2)^n$

(f) $n/3^n$

Solution. (c) is increasing; (a) and (f) are decreasing; (a), (b), (d), and (f) are bounded.

We deal with the monotonicity of the sequences first.

First, (c) is increasing because

$$(n+1)^5 = n^5 + 5n^4 + 10n^3 + 10n^2 + 5n + 1 > n^5$$

for all natural n.

Second, (a) is decreasing because

$$\frac{1}{n} > \frac{1}{n+1}$$

for all natural n (since the denominator on the left side is bigger than that on the right), and (f) is decreasing because

$$\frac{n}{3^n} = \frac{3n}{3^{n+1}} > \frac{n+1}{3^{n+1}},$$

where the inequality holds due to $3n > n+1 \iff 2n > 1$ for all natural n.

Third, (b), (d), and (e) are not monotonic. In general, to show that a sequence (s_n) isn't monotonic, it suffices to find pairs of elements (s_a, s_{a+1}) and (s_b, s_{b+1}) such that

$$s_a < s_{a+1}, \qquad s_b > s_{b+1},$$

since the first inequality shows that s_n cannot be decreasing and the second shows that it cannot be increasing. Noting this, (b) isn't monotonic because

$$-1 = \frac{(-1)^1}{1} < \frac{(-1)^2}{2} = \frac{1}{2}, \qquad \frac{1}{2} = \frac{(-1)^2}{2} > \frac{(-1)^3}{3} = -\frac{1}{3}$$

(so $a = 1$ and $b = 2$); (d) isn't monotonic because

$$\sin(6\pi/7) > \sin(7\pi/7) = 0, \qquad 0 = \sin(14\pi/7) < \sin(15\pi/7),$$

since $\sin(x)$ is positive whenever it can be written as $2\pi k + r$ where $k \in \mathbb{Z}$ and $r \in (0, \pi)$; and (f) isn't monotonic because

$$-2 = (-2)^1 < (-2)^2 = 4, \qquad 4 = (-2)^2 > (-2)^3 = -8.$$

Now, we determine which sequences are bounded.

First, (a) is bounded because $0 < 1/n \leq 1$ for all n; (b) is bounded because $-1 \leq (-1)^n/n \leq 1$ for all n (equivalently, $|(-1)^n/n| = 1/n \leq 1$ for all n); (d) is bounded because $-1 \leq \sin x < 1$ for all real x; and (f) is bounded because $0 < n/3^n < 1$, since $3^n > n$ for all n (this is easy to show by induction: it holds trivially for $n = 1$, and $3^n > n$ implies $3^{n+1} = 3 \cdot 3^n > 3n > n$ for all natural n).

It remains to show that (c) and (e) are unbounded. This follows easily from noting that n^5 and $|(-2)^n| = 2^n$ both diverge to $+\infty$. $\square$

Problem 10.2. *Prove Theorem 10.2 for bounded decreasing sequences.*

Solution. Let (s_n) be a bounded decreasing sequence. We show that

$$\lim_{n\to\infty} s_n = s$$

where $s = \inf\{s_n\}$ is the infimum of the sequence (s_n). In particular, we want to show that for all $\varepsilon > 0$, there exists an N such that $|s - s_n| < \varepsilon$ for all $n > N$.

First, there exists some m such that $|s - s_m| < \varepsilon$; this is because, if the above statement was false (i.e. $|s - s_m| \geq \varepsilon$ for all m), then we can find a greater upper bound for $\{s_n\}$ in the form of $s + \varepsilon/2$, which would contradict the fact that s is the greatest upper bound of $\{s_n\}$.

Then, since s_n is decreasing, if $k > m$, then $s \leq s_k \leq s_m$, so

$$\varepsilon = |s_m - s| > |s_k - s| = |s - s_k|$$

for all $k \geq m$, meaning we can take $N = m$ in the first paragraph. Thus, we have shown that s_n converges to s, since this bounding works for any arbitrary ε. □

Problem 10.7. *Let S be a bounded nonempty subset of $\mathbb{R}$ such that $\sup S$ is not in S. Prove that there is a sequence (s_n) of points in S such that $\lim s_n = \sup S$.*

Proof. Define $s = \sup S$. Let s_1 be an arbitrary element of S; we will then inductively define the rest of the sequence.

Given $s_1, \ldots, s_n$, define

$$d_n = \min\{s - s_1, s - s_2, s - s_3, \ldots, s - s_n\}$$

(of course, each $s - s_i$ is positive because $s > x$ for all $x \in S$). Then, let s_{n+1} be an element of S such that $s - s_{n+1} < \frac{d_n}{2}$; such an s_{n+1} exists because s is the supremum of S (otherwise we would be able to find a smaller upper bound for S, such as $s - \frac{d_n}{4}$). In fact, this implies that $d_k = s - s_k$ for all natural k, since s_{n+1} is chosen specifically so that $s - s_{n+1} < s - s_i$ for all $i \in \{1, 2, \ldots, n\}$.

I then claim that $\lim_{n\to\infty} s_n = s$. To show this, we want to prove that for each $\varepsilon > 0$, $|s - s_n| = d_n < \varepsilon$ eventually. First, we bound d_n from above:

Claim — For all $n > 1$,

$$d_n < \frac{d_1}{2^{n-1}}.$$

Proof. We prove this by induction on n, noting in general that $d_{n+1} < \frac{d_n}{2}$ by definition (recall that s_{n+1} is defined so that $s - s_{n+1} < \frac{d_n}{2}$).

The base case $n = 2$ is obvious, since

$$d_2 > \frac{d_1}{2} = \frac{d_1}{2^{2-1}}.$$

Then, if $d_n < \frac{d_1}{2^{n-1}}$,

$$d_{n+1} < \frac{d_n}{2} < \frac{d_1}{2^{n-1}} \cdot \frac{1}{2} = \frac{d_1}{2^n},$$

so the claim is proven by the principle of mathematical induction. □

This means that it suffices to show that

$$\frac{d_1}{2^{n-1}} < \varepsilon$$

is eventually true because the left-hand side is bigger than d_n. We can deduce this from the fact that, if $|a| < 1$, then $\lim_{n\to\infty} a^n = 0$, since this tells us that for all $\varepsilon' > 0$, $|a^n - 0| = |a^n|$ is eventually less than ε'. Picking $a = \frac{1}{2}$ and $\varepsilon' = \frac{\varepsilon}{2d_1}$ shows us that

$$\frac{1}{2^n} < \frac{\varepsilon}{2d_1} \iff \frac{d_1}{2^{n-1}} < \varepsilon$$

eventually.

Therefore, the limit of our (s_n) sequence is s, as desired. □

Remark. Since (s_n) is monotonically increasing and bounded by construction, the proof that $\lim s_n = s$ is really just a proof that $s = \sup\{s_n\}$, since we are just showing that there exist elements of $\{s_n\}$ that are arbitrarily close to s (so s is the least upper bound).

Problem 10.10. *Let $s_1 = 1$ and $s_{n+1} = \frac{1}{3}(s_n + 1)$ for $n \geq 1$.*

(a) *Find s_2, s_3, and s_4.*

(b) *Use induction to show that $s_n > \frac{1}{2}$ for all n.*

(c) *Show that (s_n) is a decreasing sequence.*

(d) *Show $\lim s_n$ exists and find $\lim s_n$.*

Solution. For (a),

$$s_2 = \frac{1}{3}(1+1) = \frac{2}{3}, \quad s_3 = \frac{1}{3}\left(\frac{2}{3}+1\right) = \frac{5}{9}, \quad s_4 = \frac{1}{3}\left(\frac{5}{9}+1\right) = \frac{14}{27}.$$

For (b), we see that, clearly, $s_1 = 1$, and, if $s_n > \frac{1}{2}$, then

$$s_{n+1} = \frac{1}{3}(s_n + 1) > \frac{1}{3}\left(\frac{1}{2}+1\right) = \frac{1}{2},$$

so, by the principle of mathematical induction, $s_n > \frac{1}{2}$ for all n.

To do parts (c) and (d), we will prove the following statement:

Claim — For all $n > 1$,

$$s_n = \frac{1}{2 \cdot 3^{n-1}} + \frac{1}{2}.$$

Proof. Rewrite the recurrence relation as

$$3s_n = s_{n-1} + 1 \iff 3\left(s_n - \frac{1}{2}\right) = s_{n-1} - \frac{1}{2}$$

where the second equation is a result of subtracting $\frac{3}{2}$ from both sides. Then, if (t_n) is the sequence given by $t_n = s_n - \frac{1}{2}$ for all $n \in \mathbb{N}$, we can write

$$3t_n = t_{n-1}$$

for all $n > 1$. Then, by induction, we can say that $t_n = \frac{t_1}{3^{n-1}}$: the base case $n = 1$ is obvious ($t_1 = \frac{t_1}{3^0}$), and, for the inductive step, if $t_n = \frac{t_1}{3^{n-1}}$, then

$$t_{n+1} = \frac{t_n}{3} = \frac{t_1}{3^{n-1} \cdot 3} = \frac{t_1}{3^n}.$$

Thus, we can say that, for all n,

$$s_n - \frac{1}{2} = t_n = \frac{t_1}{3^{n-1}},$$

and we know that $t_1 = s_1 - \frac{1}{2} = \frac{1}{2}$, which gives us

$$s_n = \frac{1}{2 \cdot 3^{n-1}} + \frac{1}{2}.$$

□

Now, to show (c) (that (s_n) is decreasing), we simply observe that

$$s_n - s_{n+1} = \left(\frac{1}{2 \cdot 3^{n-1}} + \frac{1}{2}\right) - \left(\frac{1}{2 \cdot 3^n} + \frac{1}{2}\right) = \frac{3}{2 \cdot 3^n} - \frac{1}{2 \cdot 3^n} = \frac{1}{3^n} > 0,$$

implying $s_{n+1} < s_n$ for all $n \geq 2$ (and obviously $\frac{2}{3} = s_2 < s_1 = 1$ is true).

Finally, for part (d), we can see very easily that the limit of s_n is $\frac{1}{2}$, since, as n goes to infinity, the limit of $\frac{1}{3^{n-1}}$ is zero.

Also, here's another (slightly quicker?) solution path for (c) and (d) based on the problem shown in recitation. For (c), induct on n to show $s_n > s_{n+1}$ for all n, with $n = 1$ giving the obvious $1 = s_1 > s_2 = \frac{2}{3}$ and $s_n > s_{n+1}$ implying

$$s_{n+1} = \frac{s_n + 1}{3} > \frac{s_{n+1} + 1}{3} = s_{n+2}.$$

For (d), we see that the limit of s_n exists because s_n is bounded (each s_n is in the interval $[\frac{1}{2}, 1]$ by the work above) and monotonically decreasing, so, letting $L = \lim s_n$, we get

$$L = \lim_{n\to\infty} s_n = \lim_{n\to\infty} \frac{s_{n-1} + 1}{3} = \frac{\lim_{n\to\infty} s_{n-1} + \lim_{n\to\infty} 1}{\lim_{n\to\infty} 3} = \frac{L+1}{3},$$

which can easily be solved to get $L = \frac{1}{2}$. $\square$

Problem 11.2, 11.3. *Consider the following eight sequences:*

$$a_n = (-1)^n, \quad b_n = \frac{1}{n}, \quad c_n = n^2, \quad d_n = \frac{6n+4}{7n-3},$$

$$s_n = \cos\left(\frac{\pi n}{3}\right), \quad t_n = \frac{3}{4n+1}, \quad u_n = \left(-\frac{1}{2}\right)^n, \quad v_n = (-1)^n + \frac{1}{n}.$$

(a) *For each sequence, give an example of a monotone subsequence.*

(b) *For each sequence, give its set of subsequential limits.*

(c) *For each sequence, give its* lim sup *and* lim inf.

(d) *Which of the sequences converges? diverges to* $+\infty$*? diverges to* $-\infty$*?*

(e) *Which of the sequences is bounded?*

Solution. **(a)** (b_n), (c_n), (d_n), and (t_n) are all already monotonic, so the sequences themselves (or any subsequence at all) suffice as answers. The only nontrivial one of these to verify is d_n; for completeness, we verify this by noting that

$$d_n = \frac{6}{7} + \frac{10}{49n-21},$$

which is decreasing because $49n - 21$ is always positive and increasing. (a_n) and (s_n) are periodic, so for example

$$(1, 1, 1, 1, \dots) = (a_2, a_4, a_6, a_8, \dots) = (s_6, s_{12}, s_{18}, s_{24}, \dots)$$

is an example of a monotone subsequence, and finally, for u_n, we can take

$$\left(\frac{1}{4}, \frac{1}{16}, \frac{1}{64}, \frac{1}{256}, \dots\right) = (u_2, u_4, u_6, u_8, \dots)$$

and for v_n we can take

$$\left(\frac{3}{2}, \frac{5}{4}, \frac{7}{6}, \frac{9}{8}, \dots\right) = (v_2, v_4, v_6, v_8, \dots).$$

(b) Big list:

- The set of subsequential limits of (a_n) is $\{-1, 1\}$.
- The set of subsequential limits of (b_n) is $\{0\}$.
- The set of subsequential limits of (c_n) is $\{+\infty\}$.
- The set of subsequential limits of (d_n) is $\{\frac{6}{7}\}$.
- The set of subsequential limits of (s_n) is $\{-1, -\frac{1}{2}, \frac{1}{2}, 1\}$.
- The set of subsequential limits of (t_n) is $\{0\}$.
- The set of subsequential limits of (u_n) is $\{0\}$.
- The set of subsequential limits of (v_n) is $\{-1, 1\}$.

(c) We can just look at the previous part's answers and compute the supremum/infimum of each set of subsequential limits:

- $\limsup a_n = 1$ and $\liminf a_n = -1$
- $\limsup b_n = 0$ and $\liminf b_n = 0$
- $\limsup c_n = +\infty$ and $\liminf c_n = +\infty$
- $\limsup d_n = \frac{6}{7}$ and $\liminf d_n = \frac{6}{7}$
- $\limsup s_n = 1$ and $\liminf s_n = -1$
- $\limsup t_n = 0$ and $\liminf t_n = 0$
- $\limsup u_n = 0$ and $\liminf u_n = 0$
- $\limsup v_n = 1$ and $\liminf v_n = -1$

(d) Convergence/divergence to $\pm\infty$ can be seen by whether the lim sup and lim inf are the same value.

- (a_n) neither converges nor diverges.
- (b_n) converges to 0.
- (c_n) diverges to $+\infty$.
- (d_n) converges to $\frac{6}{7}$.
- (s_n) neither converges nor diverges.
- (t_n) converges to 0.
- (u_n) converges to 0.
- (v_n) neither converges nor diverges.

(e) For any sequence (x_n), if both $\limsup x_n$ and $\liminf x_n$ are real numbers, then (x_n) is bounded. Thus, all the sequences except (c_n) are bounded.

$\square$

Problem 11.5. *Let (q_n) be an enumeration of all the rationals in the interval $(0, 1]$.*

(a) *Give the set of subsequential limits for (q_n).*

(b) *Give the values of $\limsup q_n$ and $\liminf q_n$.*

Solution. The answer to (a) is the interval $\boxed{[0,1]}$. Let S be the set of subsequential limits.

We start by showing that $S \subseteq [0, 1]$. In particular, for any sequence (x_i) with $0 < x_i \leq 1$ for all x_i, we show that, if $\lim x_i$ exists, then it must be in $[0, 1]$. Suppose otherwise, that is, the limit L is either less than zero or greater than 1. If $L > 1$, then $\lim_{n\to\infty} x_n = L$ means that there exists an x_i such that

$$L - x_i = |L - x_i| < L - 1$$

(where we use the fact that $x_i \leq 1 < L$ in the equality); this gives the contradiction $-x_i < -1 \iff 1 < x_i$. Similarly, if $L < 0$, then there exists an x_i such that

$$x_i - L = |x_i - L| < 0 - L$$

(using the fact that $L < 0 < x_i$), again giving a contradiction in the form $x_i < 0$.

Now, we show that $[0, 1] \subseteq S$, i.e. any $r \in [0, 1]$ is a subsequential limit. We will make use of the following lemma, which was on homework 1:

> **Lemma** (Exercise 4.11 in the textbook) — For any real a, b with $a < b$, there are infinitely many rationals between a and b.

Proof. Suppose that there are finitely many rationals in (a, b). Then, let q be the largest one. Clearly, the interval (q, b) cannot have any rational numbers in it — however, this contradicts the density of $\mathbb{Q}$ in $\mathbb{R}$, contradiction. □

Our goal is to construct a sequence of indices (n_k) such that the subsequence of (q_n) given by $(q_{n_k})_{k\in\mathbb{N}}$ converges to r. First, set $n_1 = 1$. Then, to get n_{k+1}, we will find a rational number that appears after q_{n_k} in the (q_n) sequence (i.e. $n_{k+1} > n_k$) such that $|r - q_{n_{k+1}}| < \frac{1}{2^{k+1}}$.

To show that such a rational number exists, we note that the interval

$$\left(r - \frac{1}{2^{n+1}}, r + \frac{1}{2^{n+1}}\right) \cap [0, 1]$$

contains infinitely rational numbers between 0 and 1. Since there are only finitely rational numbers appearing at or before q_{n_k} in the (q_n) sequence (namely $q_1, \ldots, q_{n_k}$), there must exist a rational number $q \in I$ such that the natural number m for which $q_m = q$ is greater than n_k; set $n_{k+1} = m$.

In this way, we constructed an increasing sequence (n_k) of naturals, which induces the subsequence (q_{n_k}) of (q_n). Moreover, we know that, for each $\varepsilon > 0$, there exists a K such that $\varepsilon > \frac{1}{2^k}$ for all $k > K$, so indeed

$$|q_{n_k} - r| < \frac{1}{2^k} < \varepsilon$$

for $k > K$, hence the limit of the (q_{n_k}) subsequence is indeed r, as desired.

Since S and $[0, 1]$ contain each other, they are the same set, which is what we claimed.

Part (b) is then obvious, since $\limsup q_n$ and $\liminf q_n$ are respectively $\sup[0, 1] = 1$ and $\inf[0, 1] = 0$. □

Problem 11.8. *Prove that* $\liminf s_n = -\limsup(-s_n)$ *for any sequence* (s_n).

Solution. We use this lemma relating supremums and infimums (suprema and infima?):

> **Lemma** (Exercise 5.4 in the textbook) — For any set S, $\inf S = -\sup(-S)$, where $-S$ is the set $\{-s : s \in S\}$.

Proof. By definition, $c = \inf S$ is the greatest lower bound of S. Since $c \leq x$ for all $x \in S$, $-c \geq -x$ for all $x \in S$, or equivalently $-c \geq y$ for all $y \in -S$, so $-c$ is an upper bound for $-S$, in particular $-c \geq \sup(-S)$.

Suppose for contradiction that $-c > \sup(-S)$. Then, there exists an ε for which every element of $-S$ is at least ε away from $-c$; that is, $|-c-y| \geq \varepsilon$ for all $y \in -S$. However, this means that $|(-y)-c| \geq \varepsilon$ for all $y \in -S$, or equivalently

$$|x - c| \geq \varepsilon \text{ for all } x \in S. \tag{1}$$

Now, since $c = \inf S$, we know that, for every $\hat{\varepsilon}$, there exists an $x_{\hat{\varepsilon}} \in S$ such that

$$|x_{\hat{\varepsilon}} - c| < \hat{\varepsilon}.$$

This gives the desired contradiction by taking $\hat{\varepsilon} = \varepsilon$ and $x = x_\varepsilon$ in (1). □

To that end, we have

$$\liminf s_n = \lim_{n\to\infty} \inf\{s_k : k \geq n\} = \lim_{n\to\infty} -\sup\{-s_k : k \geq n\} = -\lim_{n\to\infty} \sup\{-s_k : k \geq n\}.$$

The latter expression is then $-\limsup(-s_n)$, so $\liminf s_n = -\limsup(-s_n)$ as desired. □